

AI-GARCH-M

An Asymmetric Indicator GARCH-in-Mean Model for Tail-Event Risk Premia

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What is volatility?

Introduction

- Volatility is the conditional standard deviation of returns, the **price of risk** and the central object in pricing, hedging and risk management.
- It is **not observed** directly, it must be modelled from the return series.
- Returns themselves look unpredictable, but their **magnitude** is highly predictable. That predictability is what GARCH captures.

Stylised facts a volatility model must respect

- **Autocorrelation.** Returns are nearly uncorrelated, but squared and absolute returns are strongly, slowly-decaying autocorrelated.
- **Volatility clustering.** Large moves follow large moves, calm follows calm.
- **Risk premium.** Higher conditional variance is compensated by higher expected return, weakly and positively.
- **Asymmetry / leverage.** Negative shocks raise future risk more than positive shocks of equal size.
- **Heavy tails.** Standardised returns are fat-tailed, extreme moves are far more frequent than the Gaussian implies.

Literature

- **The GARCH lineage.** Engle (1982, ARCH) and Bollerslev (1986, GARCH) model the variance process. Engle, Lilien & Robins (1987) put the risk premium **in the mean**: GARCH-M.
- **Asymmetry.** Black (1976) documents the leverage effect, formalised later by GJR (1993) and EGARCH (Nelson, 1991).
- **Extreme volatility.** Granger and co-authors on long memory and extreme moves, and the GARCH-EVT strand grafting extreme value theory onto residuals.
- **Our work stays inside the GARCH framework** and adds one new piece: an EVT-identified **tail-event premium in the mean**.

The model

What we propose: AI-GARCH-M

$$r_t = \mu + \lambda_1 \sigma_{t-1}^2 + \lambda_2 I_{t-1} \sigma_{t-1}^2 + \lambda_\kappa \tilde{I}_{t-1} \sigma_{t-1}^2 + \varepsilon_t$$
$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \varepsilon_t = \sigma_t \xi_t$$

■ M0 base ■ $+\lambda_1$ M1 ■ $+\lambda_\kappa$ M2 ■ $+\lambda_2$ M3

- Leverage indicator $I_t = \mathbf{1}\{\varepsilon_t < 0\}$.
- **Tail indicator** $\tilde{I}_t = \mathbf{1}\{\xi_t < \kappa\}$, fires when the standardised innovation crosses a deep-tail threshold κ .
- The asymmetry premium is carried in the **mean** (λ_2), not in a GJR variance term.
- **Innovations.** ξ_t i.i.d. with $E[\xi_t] = 0$, $\text{Var}[\xi_t] = 1$ (so σ_t^2 is the conditional variance), and finite higher moments as needed ($E[\xi_t^4] < \infty$ for the variance results).

Parameters: what we expect

Param	Role	Expected sign	Magnitude
μ	mean	≈ 0	tiny
λ_1	GARCH-M risk premium	+ (risk-return)	small
λ_2	asymmetry premium	+ (after down moves)	small
λ_κ	tail-event premium	+, largest	sim truth 0.50
ω	variance intercept	+	small
α	ARCH	+	≈ 0.05
β	GARCH persistence	+	≈ 0.85

Stationarity: $\alpha + \beta < 1$ (≈ 0.90 persistence). κ is the q -quantile of the innovation law, $q \in \{2.5\%, 5\%\}$.

Nested hierarchy

Each model adds one term to the one before it:

$$\mathbf{M0} = \mu + \varepsilon_t \quad (\text{plain GARCH})$$

$$\mathbf{M1} = \mathbf{M0} + \lambda_1 \sigma_{t-1}^2 \quad (\text{risk premium})$$

$$\mathbf{M2} = \mathbf{M1} + \lambda_\kappa \tilde{l}_{t-1} \sigma_{t-1}^2 \quad (\text{tail-event premium})$$

$$\mathbf{M3} = \mathbf{M2} + \lambda_2 l_{t-1} \sigma_{t-1}^2 \quad (\text{asymmetry — full model})$$

Model selection by **AIC** throughout (the GARCH family is misspecified for the tails).

1. **Theory first.** Derive stationarity and closed-form moments ($E[\sigma^2]$, $E[\sigma^4]$, $\text{Var}[r]$ as functions of the parameters and the tail probability $p_\kappa = P(\xi < \kappa)$); full derivations in the appendix.
2. **Simulation.** A Monte Carlo grid checks that the parameters, and λ_κ in particular, are recoverable.
3. **Data.** Take the validated machinery to the S&P 500.

Simulation and first S&P fit

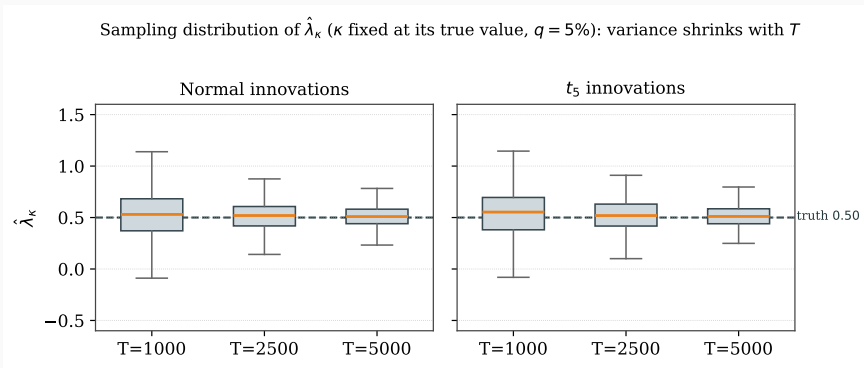
A full-factorial grid, $2 \times 3 \times 2 \times 3 = 36$ cells, $\times 1000$ replications:

Factor	Levels
Innovation law	Normal, t_5
Tail premium (truth)	$\lambda_\kappa \in \{0, 0.25, 0.50\}$
Threshold	$q^* \in \{2.5\%, 5\%\}$
Sample size	$T \in \{1000, 2500, 5000\}$

- $\lambda_\kappa = 0$ is the null cell (size of the test); 0.25, 0.50 probe recovery and power.
- Each replication fits M0–M3 by a single L-BFGS-B with restarts.

Simulation results

- With κ fixed at its true value, λ_κ at truth 0.50 **recovers cleanly**: mean 0.506–0.518, RMSE 0.10–0.15.
- Conditional- σ path recovered essentially exactly under Normal (corr 0.9995) and tightly under t_5 (corr 0.967).
- Parameters are $\sqrt{T}$ -consistent across the grid.



S&P 500 with a static κ

Fit M0–M3 with κ fixed at the **Gaussian q -quantile** (the naive choice: -1.96 , -1.64). **M3 is AIC-best.**

	Model	k	$\log L$	AIC	BIC
M0	GARCH(1,1)	4	-9107.90	18223.79	18250.99
M1	GARCH-M	5	-9105.34	18220.68	18254.67
	M3 ⁻ (no κ)	6	-9098.38	18208.77	18249.56
M2	+ λ_{κ} ($q=2.5\%$)	6	-9102.28	18216.55	18257.35
M2	+ λ_{κ} ($q=5\%$)	6	-9102.83	18217.66	18258.45
M3	AI-GARCH-M ($q=2.5\%$)	7	-9096.16	18206.31	18253.91
M3	AI-GARCH-M ($q=5\%$)	7	-9095.47	18204.95	18252.54

S&P 500 daily, $N = 6626$. $\Delta\text{AIC} = -3.8$ for M3 vs M3⁻ (AIC-best); M3⁻ is BIC-best.

M3⁻ (no κ) = the full model minus the tail term — asymmetry in the *mean*, plain GARCH variance:

$$r_t = \mu + \lambda_1 \sigma_{t-1}^2 + \lambda_2 I_{\{\varepsilon < 0\}} \sigma_{t-1}^2 + \varepsilon_t.$$

Break

Estimating κ : the idea

Estimating κ

- So far κ was **fixed** at a known quantile. On real data it is unknown, and it lives in the **far tail** where samples are scarce, so a raw empirical quantile is noisy.
- We deliberately do **not** let the in-mean fit choose its own κ : that would let the premium fish for the tail that flatters it.
- Instead, estimate κ **separately** with extreme value theory, then plug $\hat{\kappa}$ into the model. Two steps:
 - **GEV** on block minima — learn the tail *shape*;
 - **GPD** on threshold exceedances — read off the *quantile* $\hat{\kappa}_q$.

Extreme value theory in one slide

EVT is the **central limit theorem for extremes**: two classical limits, each collapsing onto a **single** universal shape γ (the tail index).

- **Block maxima** $\rightarrow$ **GEV**. Take the worst move in each block (say, each month). As blocks grow, that maximum can converge to only **one** family — the Generalised Extreme Value law (Fisher–Tippett–Gnedenko).
- **Exceedances** $\rightarrow$ **GPD**. Keep only the moves beyond a high threshold u and measure how far they overshoot. The overshoot can converge to only **one** family — the Generalised Pareto distribution (Pickands–Balkema–de Haan).
- One number runs both: $\gamma > 0$ heavy (power-law) tail, $\gamma = 0$ exponential, $\gamma < 0$ bounded.
- Why we need it: κ sits in the **far tail**, where the sample is too thin to count quantiles directly. EVT fits a tail *model* and extrapolates where the histogram runs out of data.

GEV and GPD: one shape, two lenses

The block-maxima law (GEV) and the exceedance law (GPD) describe the **same tail**, and they carry the **same shape** γ . That is a theorem, not a coincidence (Fisher–Tippett $\leftrightarrow$ Pickands–Balkema–de Haan).

- **GEV** — monthly block minima of ξ : $G_\gamma(z) = \exp\{-[1 + \gamma z]_+^{-1/\gamma}\}$.
- **GPD** — exceedances beyond a threshold u : $H_{\gamma,\beta}(y) = 1 - (1 + \frac{\gamma}{\beta}y)^{-1/\gamma}$; its quantile is the number we want, $\hat{\kappa}_q$.

γ is the **high-leverage** parameter — it sets how fast the tail decays, so the far quantile $\hat{\kappa}_q$ is far more sensitive to an error in γ than in the scale β — and it is the **hardest** thing to pin down from a handful of tail points. Because γ is **shared**, we can estimate it the robust way and reuse it.

Why chain GEV $\rightarrow$ GPD, not just GPD?

A single GPD fit estimates **both** γ and β from the few exceedances. The fragile direction is γ , and its noise feeds straight into $\hat{\kappa}_q$. So we **regularise** the shape instead of re-fitting it:

- Pin $\gamma := \hat{\gamma}_{\text{GEV}}$ from the block-maxima fit — an **independent** read on the shape, from a different slice of the data — then fit **only** the easy scale β , and read off

$$\hat{\kappa}_q = u - \frac{\beta}{\gamma} \left[(q/\zeta_u)^{-\gamma} - 1 \right], \quad \zeta_u = \hat{P}(\xi < u).$$

- One fewer free parameter in the fragile direction is a deliberate **bias–variance trade**: a little bias if $\hat{\gamma}$ is off, a large variance cut on $\hat{\kappa}_q$.
- **Simulation confirms it.** Against a raw empirical tail quantile, the chained $\hat{\kappa}_q$ cuts RMSE $\sim 20\text{--}25\%$ at $q = 1\%$ and $\sim 18\text{--}24\%$ at $q = 2.5\%$ — **biggest where the tail is deepest**, exactly where κ lives. Above the POT cutoff the two coincide.

Pipeline: back to the S&P

Now estimate κ with the GEV→GPD pipeline instead of fixing it.

	Model	k	$\log L$	AIC	BIC
M0	GARCH(1,1)	4	−9107.90	18223.79	18250.99
M1	GARCH-M	5	−9105.34	18220.68	18254.67
	M3 [−] (<i>no</i> κ)	6	−9098.38	18208.77	18249.56
M2	+ λ_{κ} ($q=2.5\%$)	6	−9102.08	18216.17	18256.96
M2	+ λ_{κ} ($q=5\%$)	6	−9101.67	18215.33	18256.13
M3	AI-GARCH-M ($q=2.5\%$)	7	−9096.64	18207.27	18254.86
M3	AI-GARCH-M ($q=5\%$)	7	−9096.45	18206.89	18254.49

M3 still wins. The pipeline $\hat{\kappa}_q$ (−2.23, −1.77) is the heavy-tail-aware threshold — deeper than the naive Gaussian one, and it lands **essentially on the empirical** residual quantile. So on the S&P the EVT step buys a principled κ , not a different answer; its RMSE edge over a raw quantile is a *simulation* result, sharpest at deeper tails.

Coefficients: what the pipeline fit shows

	M0 GARCH	M1 GARCH-M	M3 [−] <i>no κ</i>	M3 ($q=2.5\%$) AI-GARCH-M	M3 ($q=5\%$) AI-GARCH-M
μ	+0.063***	+0.042***	+0.042***	+0.044***	+0.044***
λ_1	−	+0.028***	−0.008	−0.010	−0.010
λ_2	−	−	+0.073***	+0.072***	+0.067***
λ_κ	−	−	−	+0.094	+0.107**
ω	+0.025***	+0.025***	+0.025***	+0.025***	+0.025***
α	+0.121***	+0.121***	+0.120***	+0.121***	+0.120***
β	+0.860***	+0.860***	+0.861***	+0.860***	+0.861***

Newey–West HAC SEs; *** $p < .01$, ** $p < .05$, * $p < .10$. The GARCH core (ω, α, β) and μ are sharply estimated everywhere and signs match the theory. λ_1 **flips sign** once asymmetry enters — the risk- and down-move premia are collinear, and the asymmetry-in-mean term λ_2 absorbs it while staying strongly positive.

λ_κ is the lone **marginal** coefficient — HAC-significant at $q=5\%$ but not at $q=2.5\%$, and a one-parameter effect on a single realised history, so read it as suggestive.

Takeaways

- A parsimonious, **EVT-identified tail-event premium** inside the GARCH-M family.
- **AIC-preferred** over the no- κ benchmark ($M3^-$) and cleanly recoverable in simulation.
- Honest limit: underpowered as a one-parameter significance claim on a single realised history. Next: longer samples / pooled cross-sections for power, heavier-tail robustness.

Thank you.

Appendix — stationarity and existence

Variance recursion $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, $\varepsilon_t = \sigma_t \xi_t$, with $E[\xi_t] = 0$, $E[\xi_t^2] = 1$, $m_4 := E[\xi_t^4]$.

- **Covariance stationarity:** $\alpha + \beta < 1$; then the unconditional variance is finite.
- **Fourth moment exists** iff $\alpha^2 m_4 + 2\alpha\beta + \beta^2 < 1$. For Gaussian ξ , $m_4 = 3$; for standardised t_5 , $m_4 = 9$ — the heavier tail tightens the bound.

The indicators enter only the *mean*, so they leave the stationarity of σ_t^2 untouched; they reshape the unconditional variance of r_t through the tail probability $p_\kappa = P(\xi < \kappa)$.

Appendix — closed-form moments

Taking unconditional expectations of the recursion and of its square, using $E[\varepsilon^2] = E[\sigma^2]$, $E[\varepsilon^4] = m_4 E[\sigma^4]$ and $E[\varepsilon^2 \sigma^2] = E[\sigma^4]$:

$$E[\sigma^2] = \frac{\omega}{1 - \alpha - \beta}, \quad E[\sigma^4] = \frac{\omega^2 + 2\omega(\alpha + \beta) E[\sigma^2]}{1 - \alpha^2 m_4 - 2\alpha\beta - \beta^2}.$$

The $E[\sigma^4]$ denominator is positive exactly under the fourth-moment condition on the previous slide; as $\alpha + \beta \rightarrow 1$ the level $E[\sigma^2]$ diverges, and a heavier innovation tail (larger m_4) inflates $E[\sigma^4]$.

Appendix — unconditional variance of returns

Collect the in-mean loadings into $A_{t-1} = \lambda_1 + \lambda_2 I_{t-1} + \lambda_\kappa \tilde{I}_{t-1}$, so $r_t = \mu + A_{t-1} \sigma_{t-1}^2 + \varepsilon_t$. With $P(I=1) = \frac{1}{2}$, $P(\tilde{I}=1) = p_\kappa$, and noting a tail event is always a down move ($\kappa < 0 \Rightarrow \tilde{I}I = \tilde{I}$):

$$E[A] = \lambda_1 + \frac{1}{2}\lambda_2 + p_\kappa \lambda_\kappa,$$

$$E[A^2] = \lambda_1^2 + \frac{1}{2}(\lambda_2^2 + 2\lambda_1\lambda_2) + p_\kappa(\lambda_\kappa^2 + 2\lambda_1\lambda_\kappa + 2\lambda_2\lambda_\kappa).$$

Treating the indicator block as independent of the contemporaneous σ_{t-1}^2 ,

$$\text{Var}[r] = E[A^2] E[\sigma^4] - E[A]^2 E[\sigma^2]^2 + E[\sigma^2].$$

The trailing $E[\sigma^2]$ is $\text{Var}[\varepsilon_t]$; the first two terms are $\text{Var}[A_{t-1}\sigma_{t-1}^2]$. These are the moment targets the Monte Carlo must hit.